STAT 486 Final Project
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New York City Airbnb Analysis

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INTRODUCTION

Research Question: How accurately can we predict nightly Airbnb prices in New York City during the November 2025 holiday season using property characteristics, and how does the accuracy vary when listings are first segmented into unsupervised "Market Clusters" compared to traditional borough-based groupings?

- **Focus**: November 2025 (holiday travel season)
- **Approach**: Start with exploratory data analysis, then move to machine learning models (linear regression, random forests, & clustering) to predict airbnb prices
- **Why it matters**: helps us understand what drives pricing in NYC & evaluate whether location has an effect on pricing

DATA

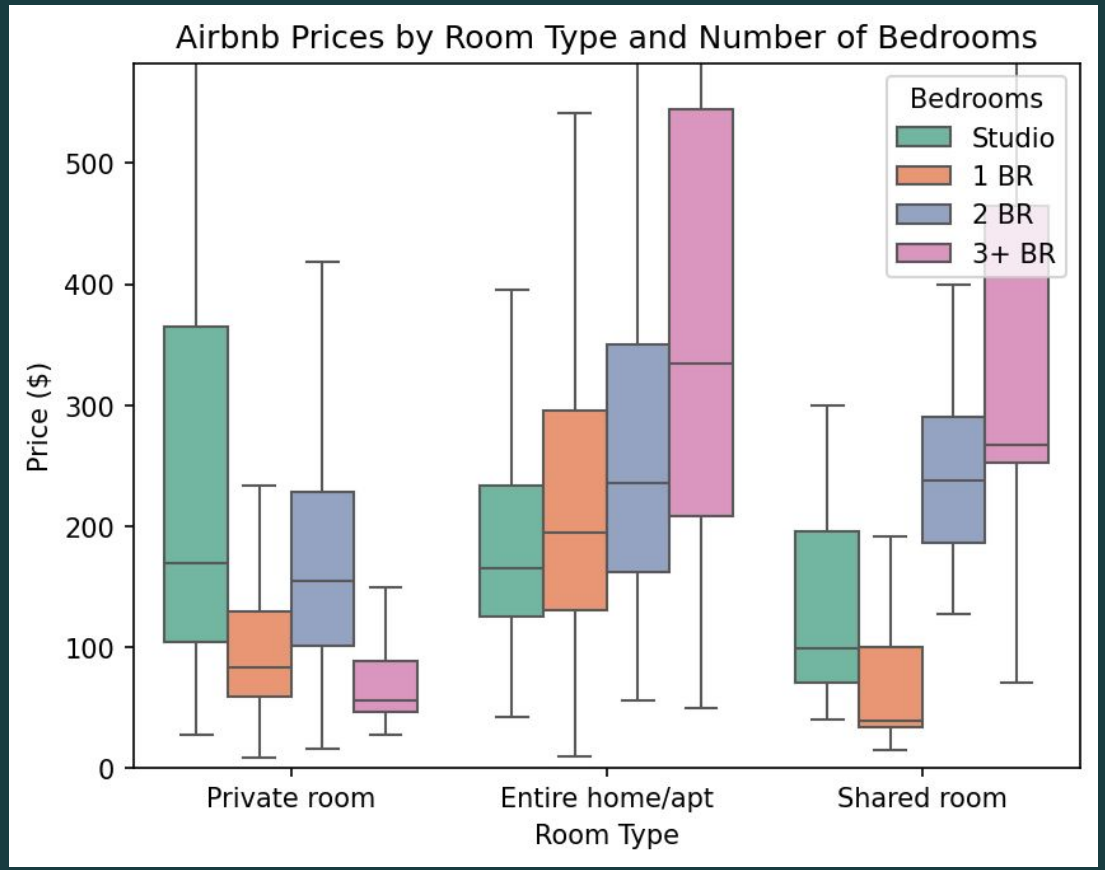
- **Data:** The NYC Airbnb listings file sourced from Inside Airbnb, a public platform that scrapes and publishes Airbnb listing data. The scrape used was November 2025.
  - 36,353 listings across NYC's five boroughs with 79 columns
- **Target Variable**: Price per night
- **Target Features**: room type, bedrooms, borough, accommodates, lat/long, & reviews.

Exploratory Data Analysis

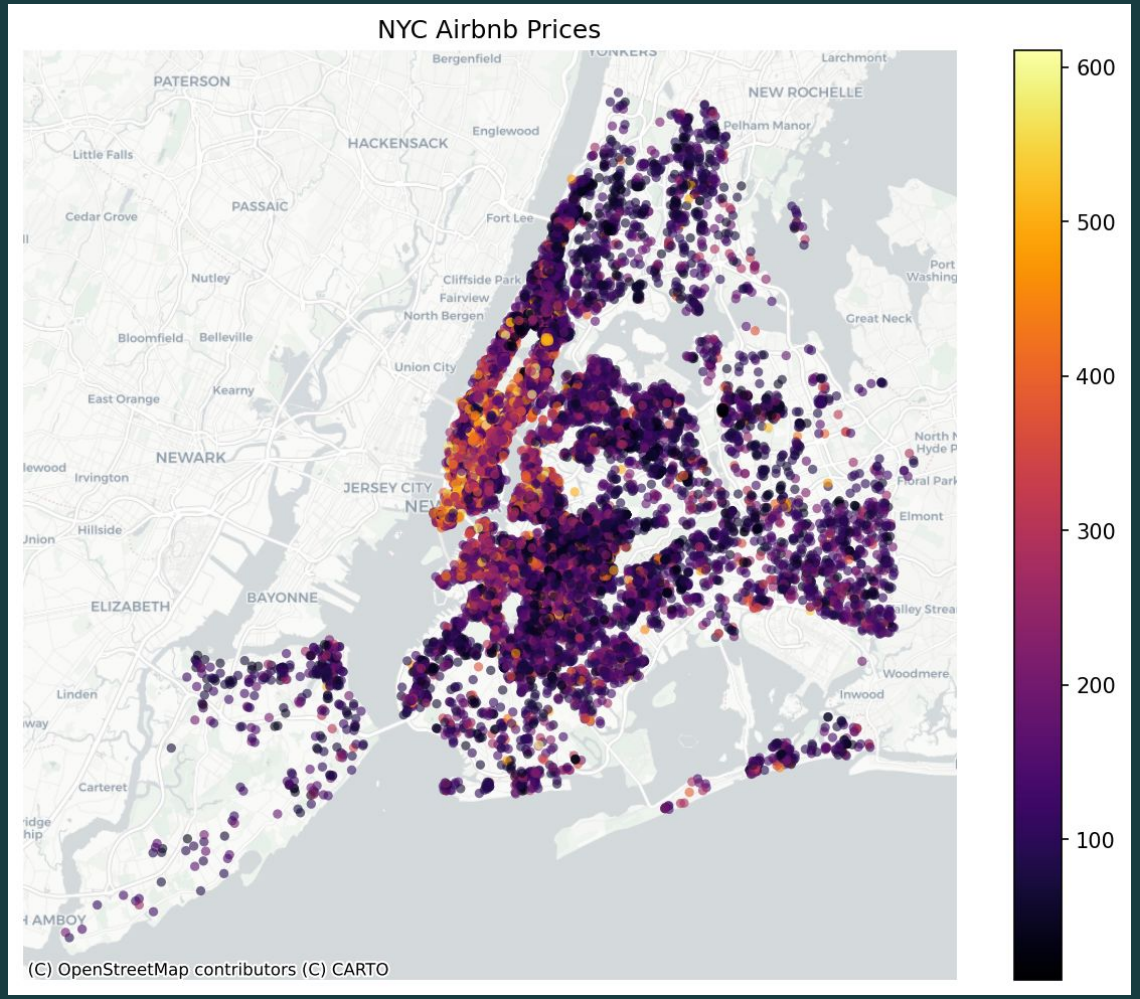
Price Distribution

Statistic	Value (USD)
Mean	$519.62
Median	$154.00
Std. Deviation	$3,658.43
25th Percentile (Q1)	$90.00
75th Percentile (Q3)	$269.00
Range	$9.00 – $50,138.00

Statistical Note: The Standard Deviation ($3,658.43) is nearly 7x the Mean, confirming high volatility. Log-transformation or outlier capping (e.g., at the 95th or 99th percentile) will be used in this analysis.



- Entire homes/apartments are consistently the most expensive
- Price increases with number of bedrooms, especially for entire units
- 1-bedroom listings show relatively stable pricing across types
- 3+ bedroom listings show high price variability



- Manhattan has the highest Airbnb prices
- Brooklyn and Queens show moderate pricing
- Listings are sparse in Staten Island, the Bronx, and outer Queens
- Prices cluster in dense, high-demand urban areas

Supervised Modeling

MODEL: Linear Regression

Predicts log-transformed Airbnb price using 9 features (bedrooms, accommodates, bathrooms, review scores, minimum nights, host listing count, room type, borough, neighbourhood). No hyperparameter tuning. Served as our baseline model.

Model explains ~62% of price variance (Test R^2 = 0.617).

Low overfitting – Train R^2 (0.643) and Test R^2 (0.617) are close, confirmed by 5-fold CV R^2 of 0.632 ± 0.022.

Test RMSE of 0.572 on log-price scale (~57% error on price).

MODEL: Decision Trees

- Tuned via GridSearchCV over max depth (3–None) and min samples per leaf (5–20).
- Best params:
  - max_depth=12
  - min_samples_leaf=5
  - Captures non-linear relationships between listing features and price.

Significant improvement over Linear Regression — Test R^2 jumped from 0.617 to 0.713, and RMSE dropped from 0.572 to 0.495.

Signs of overfitting — Train R^2 (0.798) noticeably higher than Test R^2 (0.713), suggesting the tree memorizes some training patterns.

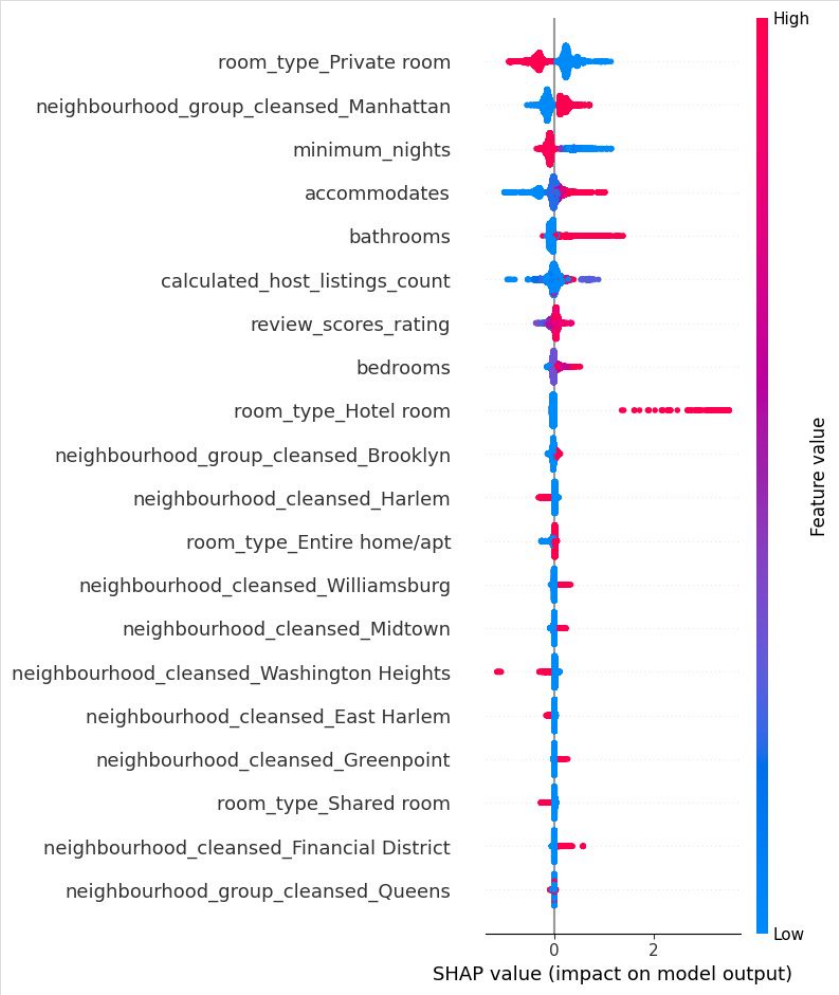
Deeper trees (max_depth=12) outperformed shallow ones, indicating price is driven by complex, hierarchical feature interactions.

MODEL: Random Forests

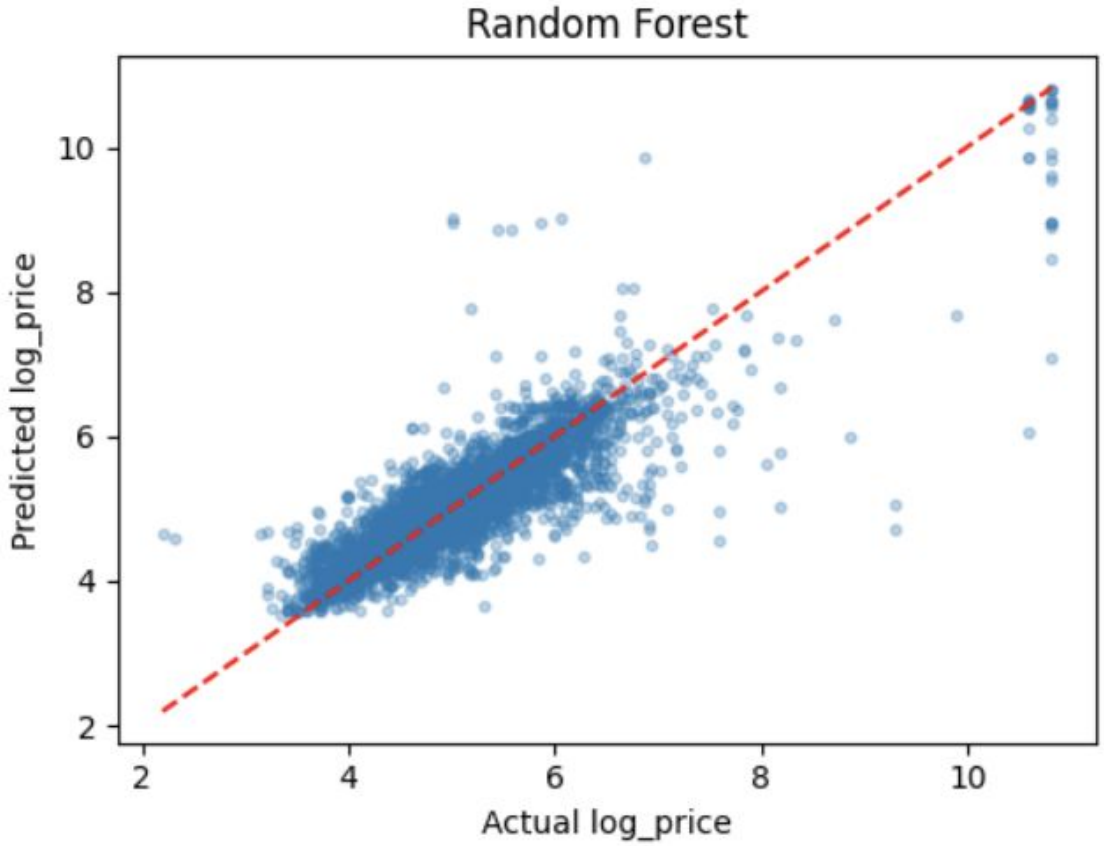
- Ensemble of 100 decision trees tuned via GridSearchCV over n_estimators, max_depth, and min_samples_leaf.
- Best params:
  - max_depth=None,
  - min_samples_leaf=5, n_estimators=100.

Best overall performance — Test R^2 = 0.752, Test RMSE = 0.460. Outperforms both Linear Regression and Decision Tree on all metrics.

More stable generalization than the single Decision Tree — smaller gap between Train R^2 (0.846) and Test R^2 (0.752) relative to improvement gained.



Best Model

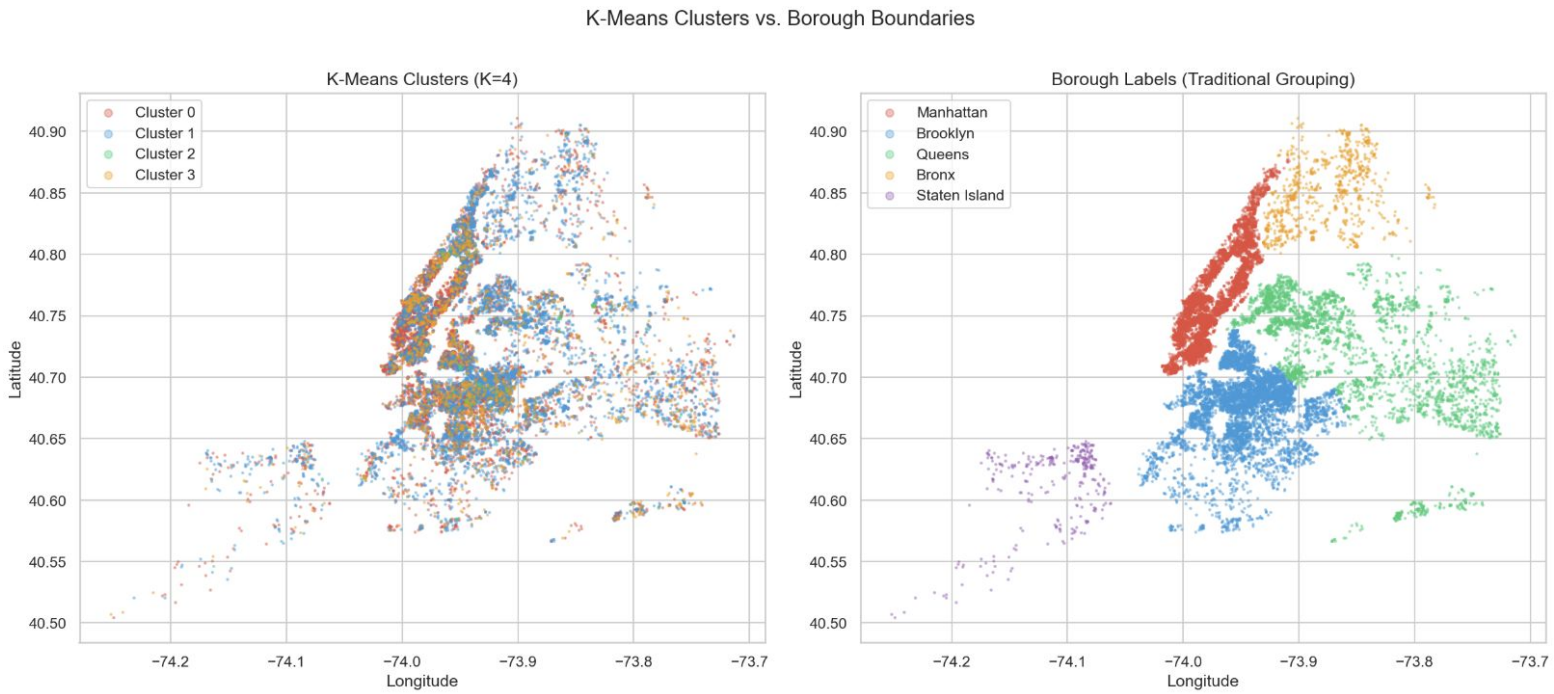

Random Forest
Predicted log_price
Actual log_price
2
4
6
8
10
2
4
6
8
10

Unsupervised Modeling

MODEL: K-Means Clustering

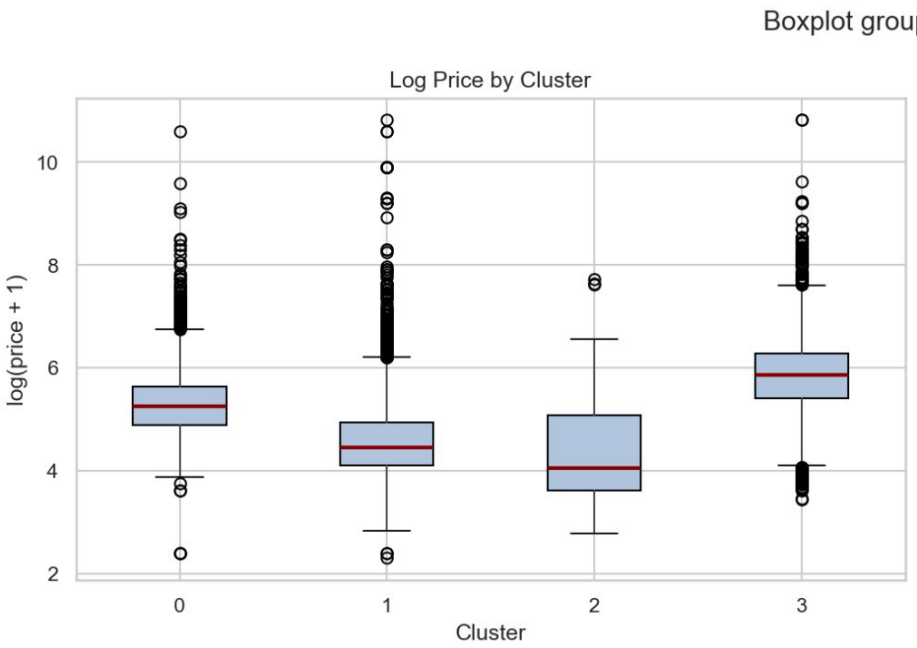
- Applied K-Means to 6 features:
  - latitude, longitude, accommodates, bedrooms, bathrooms, and room type (one-hot encoded).
- All features standardized. Optimal K=4 selected via elbow method and silhouette scores across K=2–10.

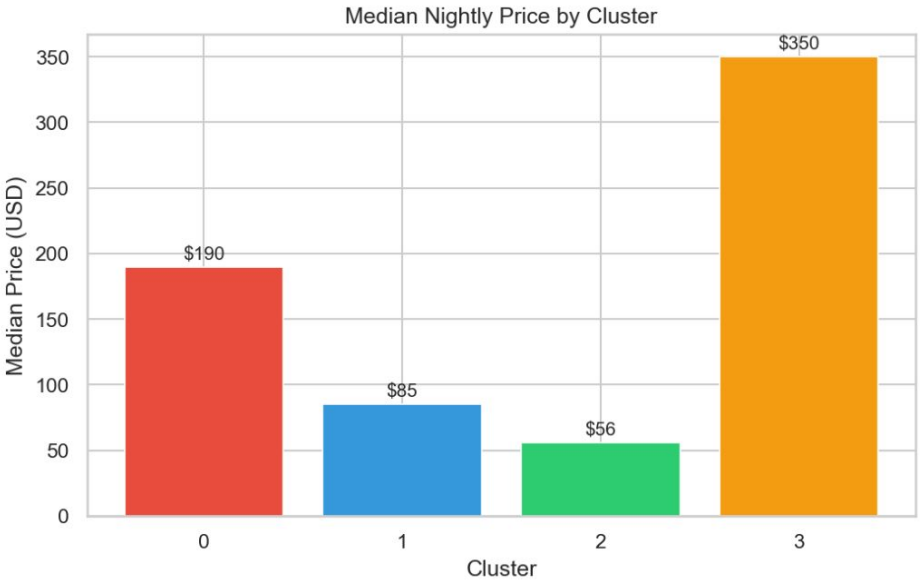
- We initially expected clusters to align with boroughs (geographic regions). However, visual comparison shows that K-Means clusters do not match borough boundaries well.



Instead, clusters align strongly with room type and price tiers, revealing meaningful market segments:

- Cluster 1: Low-cost private rooms (median ≈ $85)
- Cluster 0: Mid-range entire homes (median ≈ $190)
- Cluster 3: Large, high-end listings (median ≈ $350)
- Cluster 2: Small shared/low-demand listings





Key Takeaways

	Test R Squared	Test RMSE	Key Hyperparameters Tuned	Best Params
LINEAR REGRESSION	0.617	0.572	Features used:	
RANDOM FORESTS	0.752	0.460	• n_estimators, max_depth, min_samples_leaf	• 100 trees, depth=None, leaf=5
DECISION TREE	0.713	0.495	• Max_depth, min_samples_leaf	• depth=12, leaf=5
K-Means Clustering	Pre cluster: 0.6955 Post cluster: 0.6916	Pre: 0.4405 Post: 0.4433	latitude, longitude, accommodates, bedrooms, bathrooms, and room type (one-hot encoded).	Optimal K=4 selected via elbow method and silhouette scores across K=2–10.

Thank you